EncoderX
TECH INTERNSHIPS

Empowering Future Tech Professionals

✉ encoderxtech@gmail.com
⚑ Pakistan

TASK BOOKLET – FULL STACK DEVELOPMENT

Program: EncoderX Remote Internship Batch 02

Track: Full Stack Development

Week: 01

LEARN
BUILD
INNOVATE
SUCCEED

ABOUT ENCODERX

EncoderX provides industry-oriented remote internship experiences designed to help students build practical skills, develop professional portfolios, and gain real-world project exposure.

DO'S

✓ Maintain professional communication.
✓ Use GitHub for version control.
✓ Document your code and project clearly.
✓ Follow coding standards and best practices.
✓ Test your application before submission.
✓ Submit your work before deadlines.

DON'TS

✗ Do not submit copied projects.
✗ Do not share confidential project material.
✗ Avoid incomplete documentation.
✗ Do not expose passwords, API keys, or secrets.
✗ Do not ignore application errors or security issues.
✗ Do not submit untested or broken functionality.

TASK 1: TASK MANAGEMENT APP

W	D	T	A
EXPECTED DURATION	**DIFFICULTY LEVEL**	**TECHNOLOGY**	**DOMAIN**
1 Week	Beginner to Intermediate	Full Stack	Web Development

OBJECTIVE

The objective of this task is to build a full-featured CRUD (Create, Read, Update, Delete) application. Interns will learn how to design a user-friendly interface, construct robust backend APIs, manage database states, and secure application data using authentication.

TASK OVERVIEW

A Task Management App is a staple in enterprise software, showcasing an engineer's ability to synchronize frontend state with backend logic. In this project, interns will implement data persistence, user registration and login protocols, protected API routes, and seamless data-fetching routines.

The application should allow authenticated users to create, view, update, and delete their own tasks while providing a responsive and user-friendly interface.

LEARNING OUTCOMES

Upon successful completion of this task, interns will be able to:

- ✓ Architect and implement full CRUD operations.
- ✓ Secure endpoints using JWT or session-based authentication.
- ✓ Connect a frontend application to a backend REST API.
- ✓ Design and integrate a relational or NoSQL database.
- ✓ Manage application state across a modern user interface.
- ✓ Deploy applications using modern cloud hosting platforms.

INSTRUCTIONS

1

Step 1: Technology Stack Selection

Choose a modern full-stack technology stack such as:
- MERN Stack: MongoDB, Express.js, React, Node.js
- PERN Stack: PostgreSQL, Express.js, React, Node.js
- Django/Python Stack: Django, SQLite/PostgreSQL, React or HTML Templates
- Next.js Stack: Next.js, Prisma, PostgreSQL or MongoDB

2

Step 2: Database Setup & Authentication

Design and implement the database and authentication system.
- Create schemas for Users and Tasks, including required relationships.
- Implement user registration.
- Hash user passwords securely using a suitable method such as bcrypt.
- Implement login functionality.
- Generate authentication tokens using JWT or implement session-based authentication.
- Protect private routes using authentication middleware.

3

Step 3: Backend API Development

Create and document REST API endpoints for task management.
- **POST /api/tasks** – Create a task.
- **GET /api/tasks** – Fetch user-specific tasks.
- **PUT /api/tasks/:id** – Update task status or details.
- **DELETE /api/tasks/:id** – Remove a task.

Ensure that authenticated users can only access and modify their own tasks.

4

Step 4: Frontend Development

Build a responsive and user-friendly frontend interface containing:

- Login page.
- Registration page.
- Dashboard displaying the user's tasks.
- Tasks grouped by status: Todo, In Progress, and Done.
- Forms or modals for creating tasks.
- Forms or modals for editing tasks.
- Appropriate loading, success, and error states.

5

Step 5: Integration & Deployment

Integrate the frontend with the backend API and deploy the completed application.

- Connect the frontend to the API using `fetch` or `axios`.
- Verify authentication and protected routes.
- Test all CRUD operations.
- Deploy the backend using a platform such as Render or Railway.
- Deploy the frontend using a platform such as Vercel or Netlify.
- Verify that the complete application works through the live deployment.

DELIVERABLES

✓ Source Code – Frontend and Backend
✓ Live Deployment URL
✓ Database Schema Diagram
✓ Postman API Collection Documentation
✓ Professional README.md
✓ GitHub Repository Link

EVALUATION CRITERIA

Criteria	Weightage
Authentication & Security	**20%**
API Functionality & CRUD	**30%**
Database Design & Integration	**20%**
UI/UX & Responsiveness	**15%**
Deployment & Code Quality	**15%**

SUBMISSION REQUIREMENTS

✓ Upload all project files to a public GitHub repository.
✓ Maintain a clear and professional README file.
✓ Record a 3–5 minute project demonstration video.
✓ Publish a LinkedIn post describing your project and key features.
✓ Tag EncoderX in the post.
✓ Include hashtags: #EncoderX #FullStackDevelopment #Internship #LearningInPublic
✓ Submit the GitHub Repository Link, LinkedIn Post Link, and Video Link in a single PDF through the assigned submission portal.

ALL THE BEST!